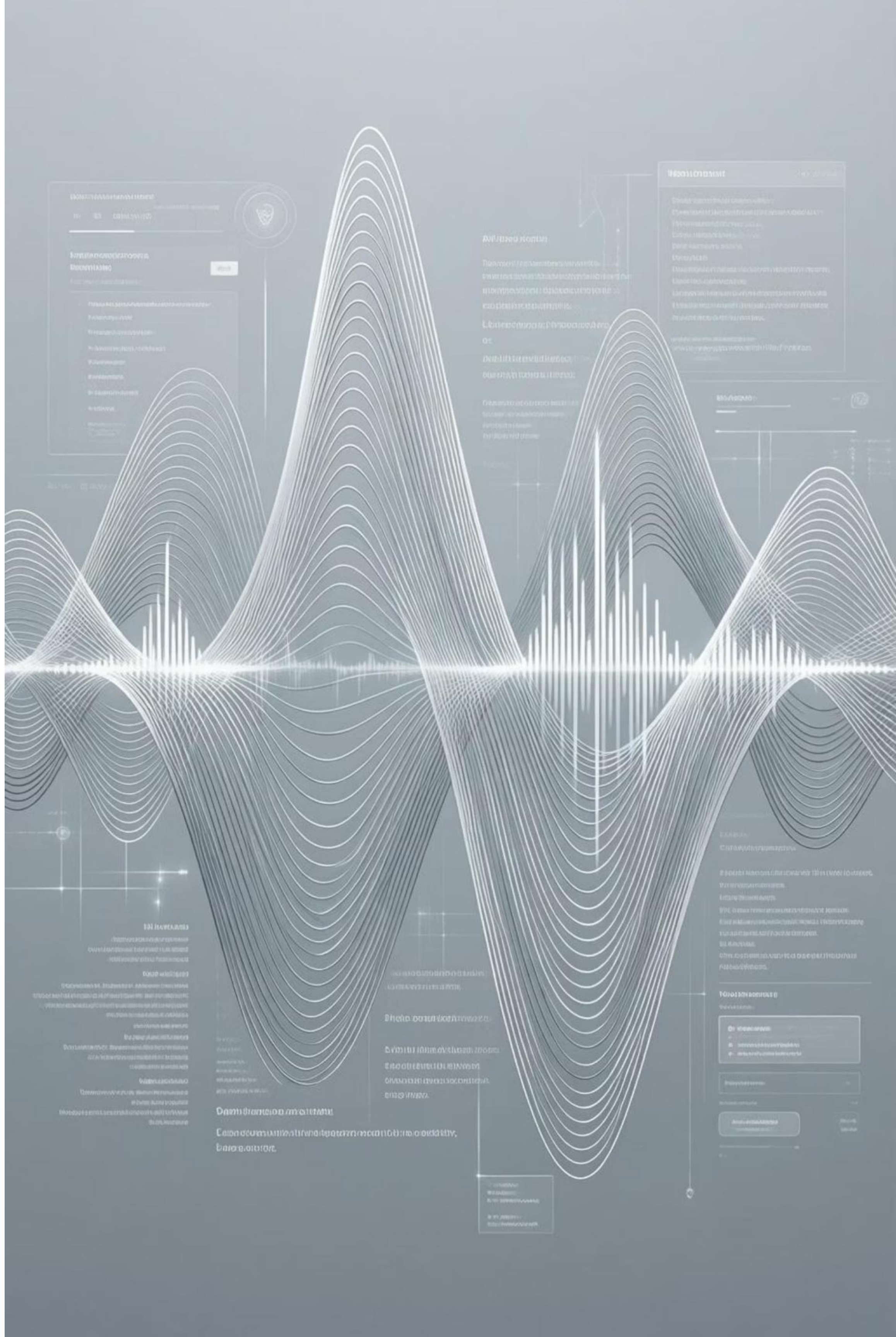
Automated Podcast Transcription and Topic Segmentation System

An AI-based solution for automatic podcast transcription and intelligent topic segmentation

-By Nidhi Paswan



Introduction to the System

In today's fast-paced digital landscape, **podcasts and lengthy audio files** have become indispensable sources of information and entertainment. However, manually sifting through hours of audio to glean insights is an **arduous and time-consuming task**. This project introduces a sophisticated automated system designed to streamline this process.

Automated Conversion

Our system automatically converts spoken content from podcast audio into accurate, readable text.

Insight Extraction

Beyond transcription, it intelligently extracts valuable insights, making audio content more accessible and actionable for users.

The Problem

The current manual approach to analyzing audio content presents several significant challenges, leading to inefficiencies and missed opportunities. We address these critical pain points to enhance user experience and productivity.

Time-Consuming Transcription

Manual transcription of podcasts and long audio files demands considerable human effort and time, often delaying content analysis.

Difficulty in Topic Identification

Identifying key topics, emotional tones, and important segments within extensive audio content is incredibly challenging without automated tools.

Lack of Analytical Tools

There is a notable absence of easy, integrated solutions for deep analysis of long-form audio, hindering efficient information retrieval.

Core Objectives

Automated Audio-to-Text Conversion

To accurately and efficiently convert spoken words from podcast audio into written transcripts.

Intelligent Topic Segmentation

To automatically divide the transcribed text into distinct and meaningful topic segments, improving navigability.

Sentiment Analysis & Summarisation

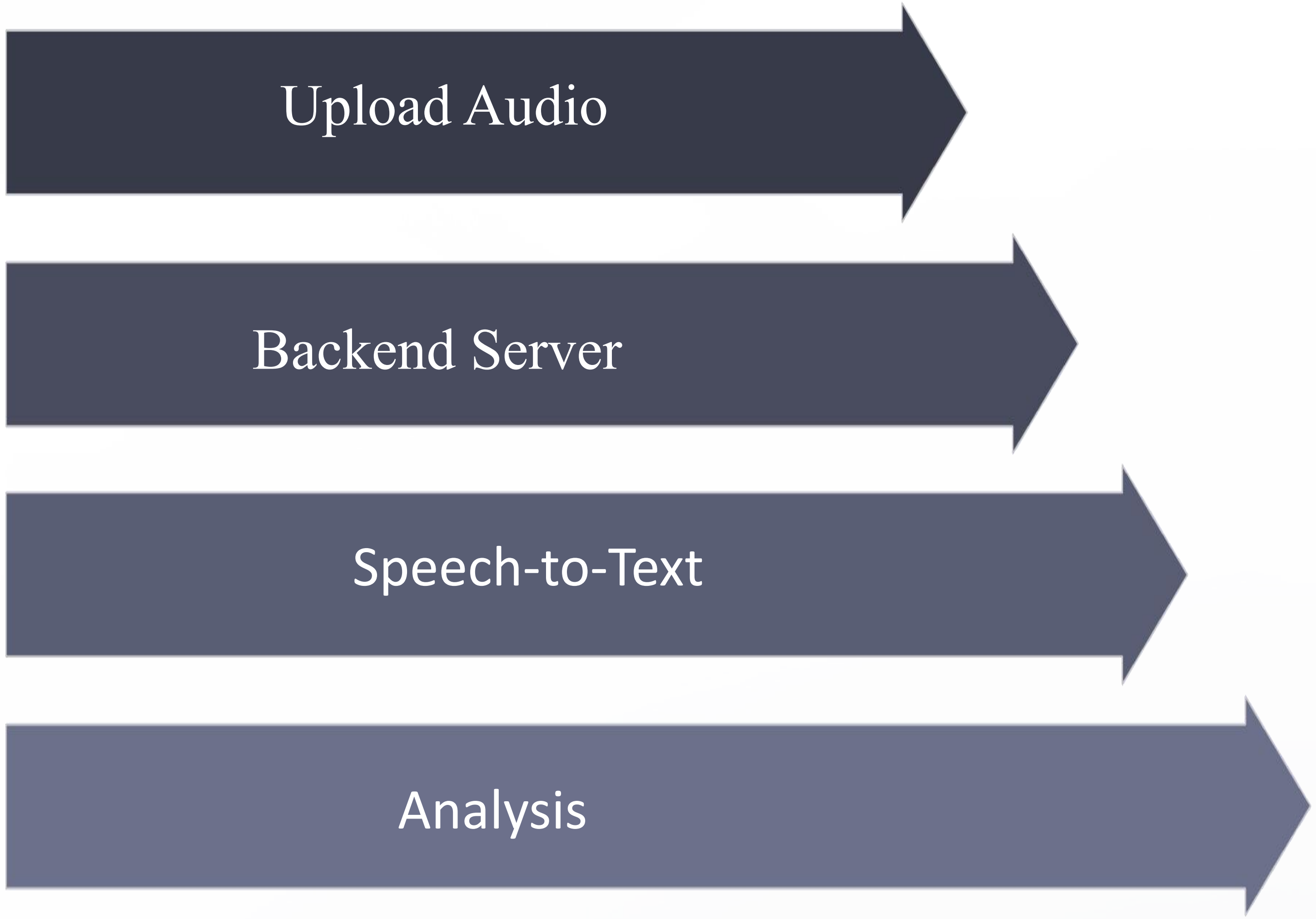
To perform sentiment analysis for emotional insights and generate concise summaries of the audio content.

Interactive User Interface Development

To provide a highly interactive, user-friendly, and responsive web interface for seamless content interaction.

System Overview: How It Works

The System follows a robust, multi-stage process to deliver comprehensive audio analysis. Each step is meticulously designed to ensure accuracy, efficiency, and an intuitive user experience.



1

User Uploads Audio

The process begins with the user uploading their desired podcast audio file through the web interface.

2

Backend Processing

The uploaded audio is securely transmitted to our robust backend server for initial processing.

3

AI-Powered Transcription

An advanced AI model converts the audio into accurate text, forming the basis for further analysis.

4

Text Analysis

The transcribed text undergoes comprehensive analysis to identify key topics and sentiments.

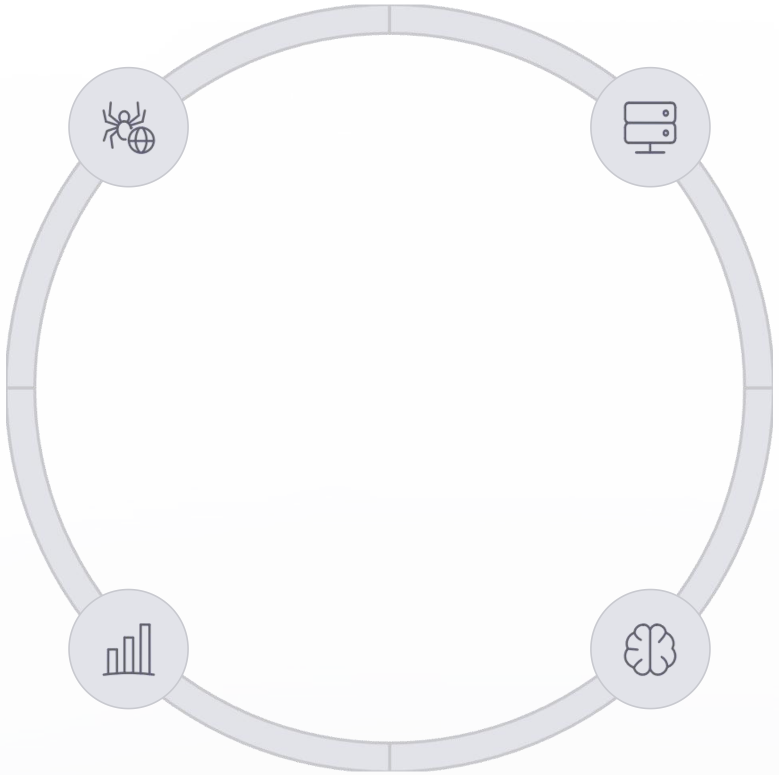
5

Results Displayed

Finally, all processed information and insights are intuitively displayed on the web interface for the user.

Technology Stack

The System leverages a powerful and modern technology stack, carefully selected to ensure scalability, performance, and a rich user experience. Each component plays a crucial role in the overall architecture, from user interaction to deep AI processing.



Frontend

React.js and **CSS** for an interactive user interface.

Backend

Python and **Flask** handling core logic and API services.

Visualization

Integrated **Charts & Graphs** for insightful data representation.

AI/ML

Whisper Model for Speech-to-Text and **NLP** for analysis.

Frontend Technologies: Building the User Experience

The user-facing part of our system is crafted using cutting-edge frontend technologies, ensuring a seamless, responsive, and visually appealing experience for all users.

React.js: Dynamic User Interface

- Developed for building a **fast and dynamic user interface**, providing a smooth interactive experience.
- Utilizes a component-based architecture for modularity, reusability, and easier maintenance.
- Operates as a Single Page Application (SPA), offering fluid transitions and improved performance.

CSS: Styling and Responsiveness

- Employs CSS (Cascading Style Sheets) for crafting a modern and aesthetically pleasing design across all elements.
- Ensures **responsive design**, making the application fully functional and visually consistent on various devices and screen sizes.
- Includes comprehensive **Dark and Light mode support**, allowing users to customize their viewing preference and reduce eye strain.

Backend Technologies: Powering the System

The robust functionality and intelligent processing of our system are powered by a meticulously chosen backend stack, providing a secure and efficient foundation for all operations.

1

CORE LOGIC

Python: Versatile & Powerful

- Python forms the backbone of our system, handling the **core backend logic** and orchestrating complex processes.
- It is instrumental in the seamless AI model integration, allowing for advanced machine learning capabilities.

2

WEB FRAMEWORK

Flask: Lightweight & Flexible

- Flask is employed for efficient REST API development, facilitating communication between frontend and backend.
- It adeptly **handles audio upload and response management**, ensuring smooth data flow for transcription.

AI FEATURES:

The Intelligence Core

The true innovation of our system lies in its sophisticated application of Artificial Intelligence and Machine Learning, which transform raw audio into actionable intelligence.

Speech-to-Text (Whisper Model)

- Utilizes the cutting-edge **Whisper Model** to accurately convert spoken language from podcast audio into precise textual transcripts.
- Engineered to **efficiently handle long audio files**, ensuring comprehensive transcription without compromising speed or accuracy.

Natural Language Processing (NLP)

- Employs robust NLP techniques for intelligent **topic segmentation**, breaking down transcripts into coherent discussion points.
- Conducts **sentiment analysis** to identify emotional tones and moods expressed within the audio content.
- Generates concise and informative **text summarization**, allowing for quick grasp of key content.

Key Features

Our Automated Podcast Transcription and Topic Segmentation System is packed with powerful features designed to make audio content more accessible, searchable, and insightful for every user.

Automatic Transcription

Converts audio to text swiftly and accurately.

Topic-Wise Segmentation

Breaks down audio into digestible, thematic sections.

Sentiment Analysis

Identifies the emotional tone of the content.

Audio Visualization

Provides a visual representation of the audio.

Searchable Transcript

Enables quick keyword searches within the text.

Audio & Text Synchronization

Connects playback to specific text segments for easy navigation.

Advantages

Saves Time & Effort

Automates the labour-intensive processes of transcription and content analysis, significantly reducing manual workload.

High Transcription Accuracy

Leverages state-of-the-art AI models, such as Whisper, to ensure highly accurate and reliable textual conversions.

Easy Analysis of Long Podcasts

Facilitates quick identification of key topics and sentiments, making complex audio content digestible and searchable.

User-Friendly Interface

Provides an intuitive and responsive web interface, ensuring a seamless and engaging experience for all users.

Applications

Podcast Platforms

Automate transcription and topic segmentation for vast audio libraries, improving searchability and listener engagement.

Online Education

Generate searchable transcripts for lectures and courses, aiding student learning and content accessibility.

Media and Journalism

Rapidly analyse interviews and broadcasts for key topics and sentiment, accelerating content creation and reporting.

Content Creators

Efficiently manage and repurpose audio content, expanding reach and improving SEO through transcribed material.

Research and Analysis

Expedite qualitative data analysis from audio sources, identifying patterns and insights for academic and market studies.

Challenges & Limitations

Hardware Constraints

The Whisper model requires substantial RAM and GPU power, posing challenges for real-time processing of very long audio files on standard hardware.

Connectivity Requirements

Certain NLP libraries and API-based modules necessitate a stable internet connection, potentially limiting usability in offline or low-bandwidth scenarios

Processing Latency

Processing extremely long audio files requires substantial computational power and can be time-consuming.

Acoustic Variability

Handling sensitive audio content necessitates stringent security and privacy protocols to protect user data.

Future Enhancements

Multi-Language Support

Enable transcription and analysis in multiple languages, significantly expanding the system's reach to a global audience.

Speaker Identification

Automatically identify and differentiate between speakers within audio, enhancing readability and analysis for multi-person podcasts.

Real-time Transcription

Develop the capability for on-the-fly transcription of live audio streams, opening up new possibilities for live events and broadcasts.

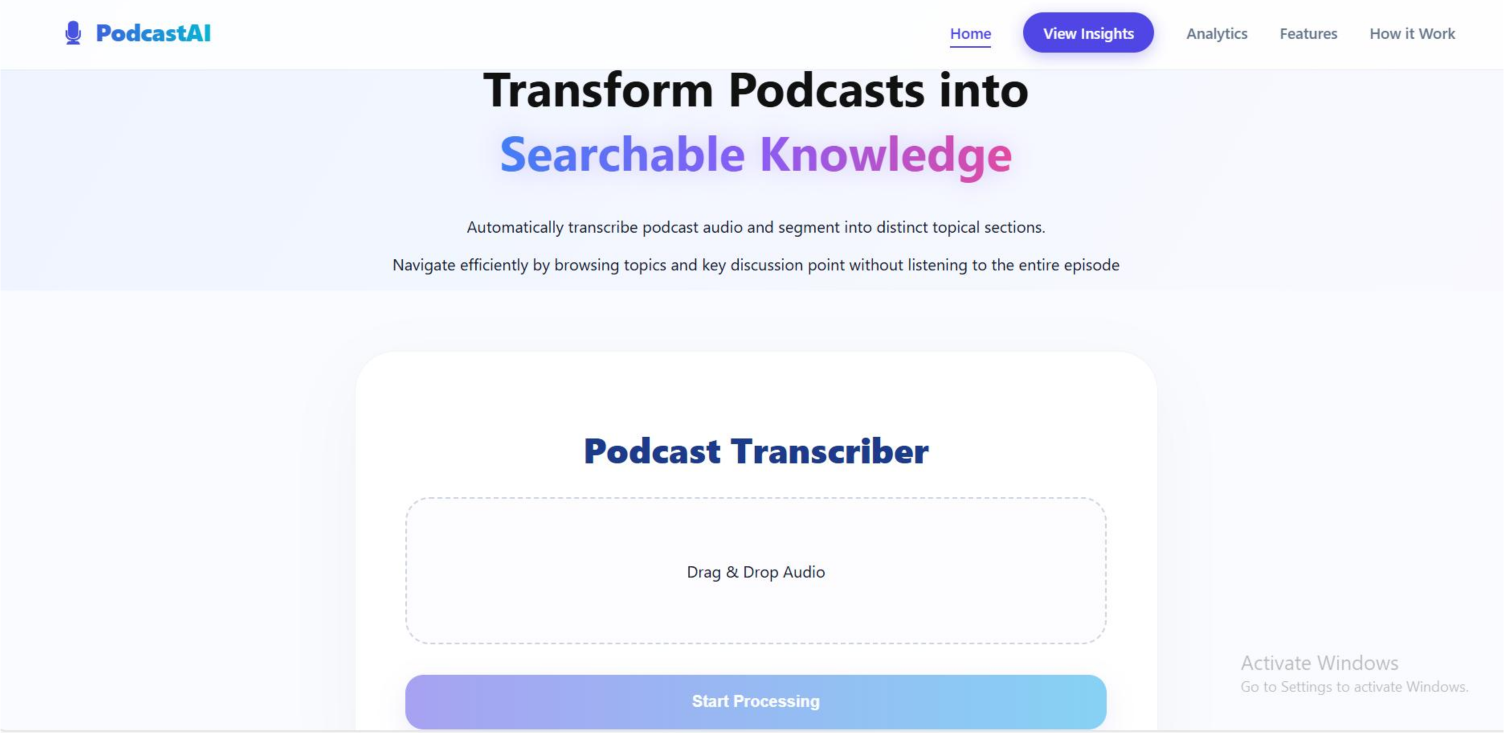
Downloadable Reports

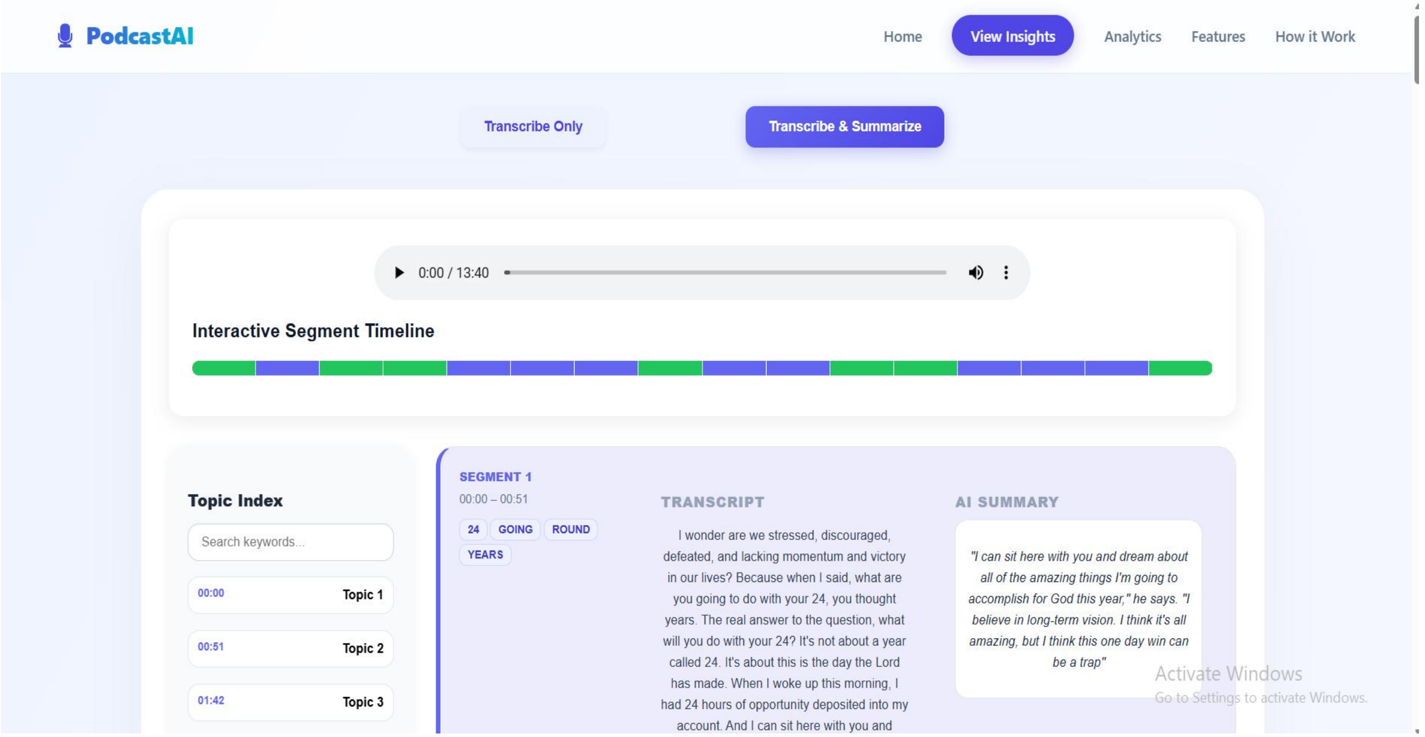
Allow users to download comprehensive reports, including full transcripts, sentiment analysis, and topic summaries in various formats (PDF, CSV).

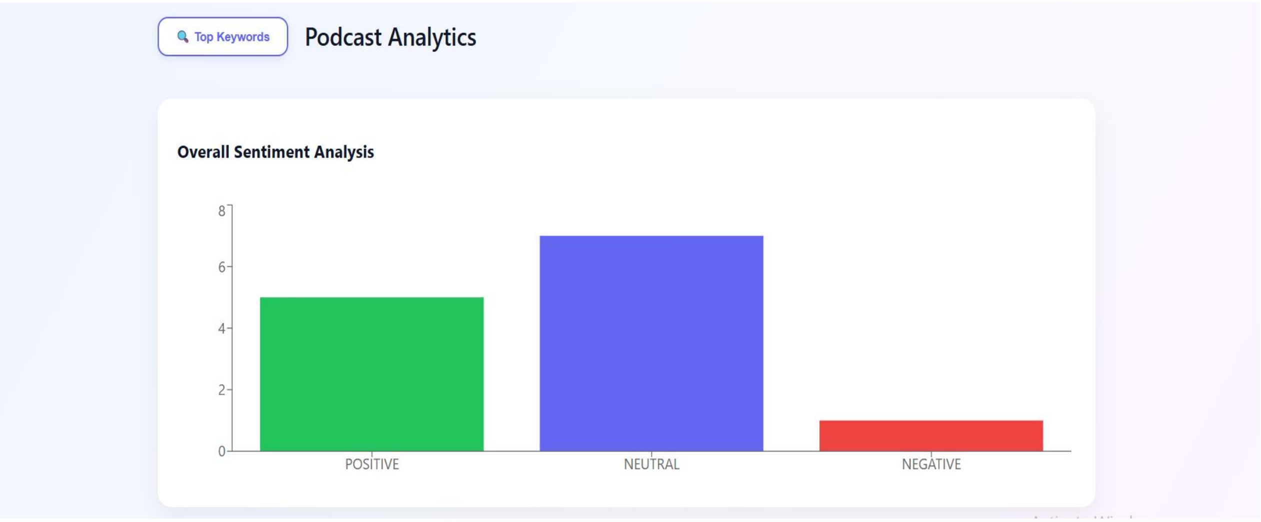
Cloud Deployment Options

Offer flexible deployment options, including cloud-based solutions, for enhanced scalability, accessibility, and ease of management.

System Landing Page







References

1. Dataset Source

The American Life Podcast Dataset and Kaggle Podcast Transcripts

2. AI Models

- **OpenAI Whisper for ASR**
- **BART** for Abstractive Summarization
- **NLTK** – Used for text preprocessing, segmentation, and sentiment analysis
- **PyDub** – Used for audio file handling, format conversion, and preprocessing
- **WordCloud** – Used for visualizing important keywords from podcast transcripts

3. Frameworks

React (Frontend), Flask (Backend), CSS (Styling)

4. Mentorship

Delivered through the Infosys Springboard Virtual Internship Program.

Q&A

Thank You!